Mass production of *Rhizobium*

The flowchart explaining the mass production of *Rhizobium* biofertilizer is given below

Efficient N_2 fixing strain of *Rhizobium* are selected and initially grown in YEM broth for 24 hours.

↓

Inoculum level is increased from 250 ml to 1000 ml in large flasks

↓

Cultures are then grown in large fermentor for large scale production

↓

Broth is blended with powdered carrier material (peat or lignite)

↓

Packing in polythene bags 200g per bag

↓

Curing at 25°C

↓

Dispatch to farmers

Method of application of *Rhizobium* to plants

Carrier based *Rhizobium* inoculants are mixed with water to form slurry to which the seeds of plants are added (Figure 11.8). The coated seeds are dried in shade and used for sowing.

11.2.2 Phosphate Solubilizers

Several soil bacteria like *Pseudomonas* and *Bacillus* possess the ability to convert insoluble mineral phosphates into soluble form by secreting organic acids thereby making it available to plants.

For mass cultivation and inoculant preparation, the cultures are grown in Pikovaskaya broth for 7-18 days and mixed in suitable carrier like peat or lignite. After curing for a week, the inoculants are packed and made ready for use in a similar manner as *Rhizobium* inoculants.

11.2. 3 VAM

Mycorrhiza means fungus root. It describes the symbiotic association between plant and fungus. Vesicular Arbuscular Mycorrhiza (VAM) is an endomycorrhiza which is used as a fungal biofertilizer. They mobilize the soluble

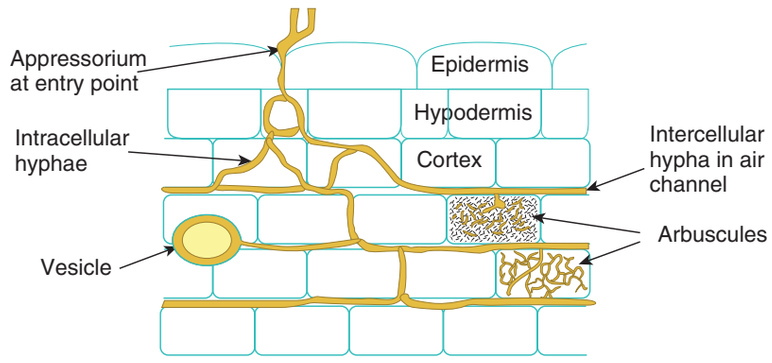


Figure 11.8: Showing the colonization of VAM fungi in root cells of plants

TN_GOVT_XI_Micro_Biology_CH11.indd 172 16-02-2019 10.04.46 AM